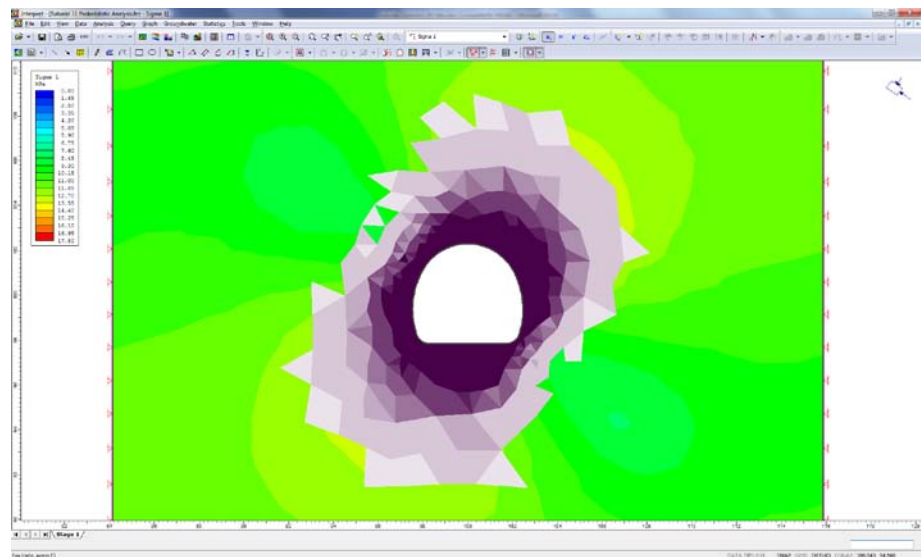


Probabilistic Analysis in *Phase2*

Often the parameters required for analysis are not well known. In these cases it is desirable to perform a parametric study where model behaviour is examined for a range of possible inputs. This tutorial describes how to do this with *Phase2*.

Topics covered

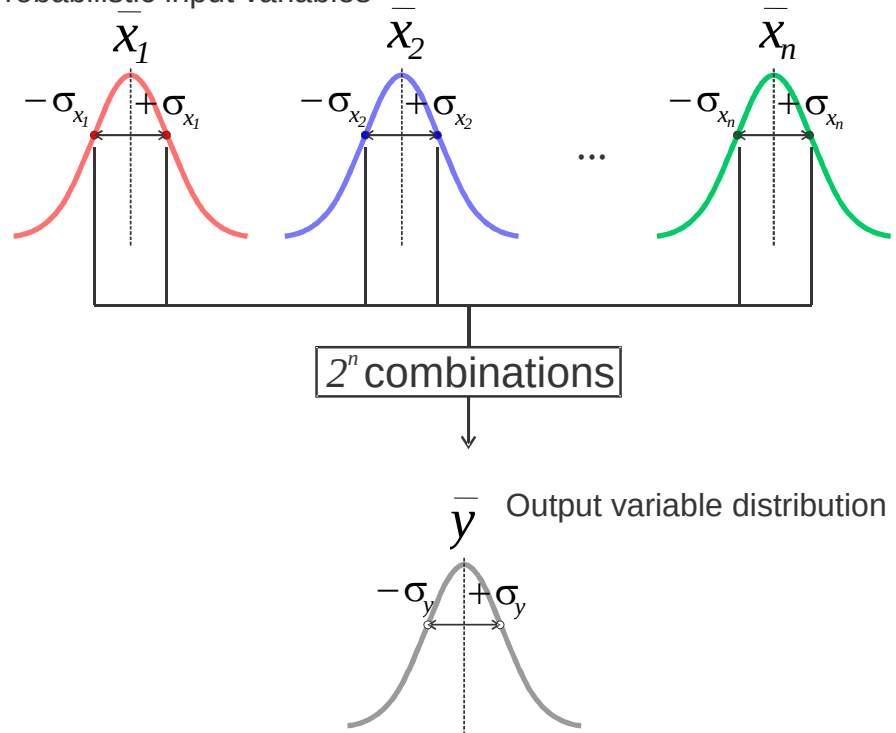
- Principles
- Statistics
- Error plots



Principles

In order to deal with probabilistic inputs and outputs in a computationally efficient method, the point estimate method (PEM) has been added to *Phase2*. The purpose of the method is to be able to combine probabilistic input variables (also called random variables) and to evaluate the distribution of the output variables. The principle of PEM is to compute solutions at various estimation points and to combine them with proper weighting in order to get an approximation of the distribution of the output variables.

Probabilistic input variables



The version of PEM implemented in *Phase2* is the two-point estimate method for the first and second moment of uncorrelated variables. The evaluation points of the random variables in a PEM analysis are located at $\pm 1\sigma$ of the mean.

The method requires 2^n evaluations of the solution, where n is the number of random variables. The distribution of the solution for $y = f(x_1, x_2, \dots, x_n)$ is given by:

$$\bar{y} = \sum_{i=1}^{2^n} w f_i \quad (1)$$

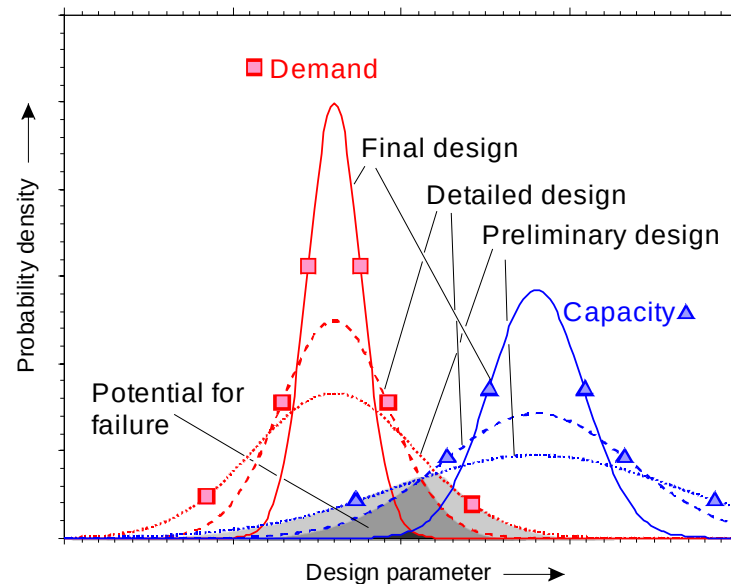
$$\sigma_y = \sqrt{\sum_i^{2^n} w f_i^2 - \left[\sum_i^{2^n} w f_i \right]^2} \quad (2)$$

where the weights w are given by $1/2^n$. f_i are successive evaluations of f at the 2^n possible combinations of the random variables at the point estimate locations, i.e. at $\bar{x}_n - \sigma_{x_n}$ and $\bar{x}_n + \sigma_{x_n}$. In the solution presented here, all input and output variables are assumed to follow a normal distribution given by their mean $\bar{x}$ and standard deviation σ_x .

Applicability

When considering statistical distributions of input parameters in geomechanics problems, three different concepts must be considered: uncertainties, variability and heterogeneities. These three concepts must be treated separately as they have various impacts on the rock mass behaviour and as different approaches must be used to tackle them.

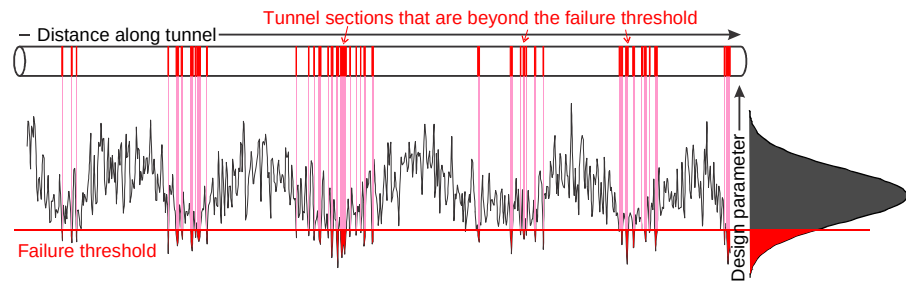
Uncertainties arise from the difficulty in measuring key geomechanical properties like rock stresses, rock modulus or rock strength. Any of these measurements is smudged with some error due to the sampling process, sample preparation or sensitivity and calibration of the measuring devices. This uncertainty is usually evaluated and reduced by acquiring repeated measurements during the development of a project.



Considering a given design criteria, the probability of failure is given by the gray areas on the Figure above: combining it with the consequences of failure allows the computation of the risk of a given design (taking the standard definition of risk being probability of occurrence times consequence of occurrence) and to evaluate if this risk is acceptable. There is always some residual risk remaining even at the final design

stage (darker grey on the Figure above). This residual risk needs to be handled by using mitigation measures, e.g. safety or worker exposure reduction procedures. The PEM method implemented in *Phase2* is adapted to handle this kind of situation, i.e. it will allow one to track how uncertainties in the input parameters are propagated through the analyses and results in uncertainty in the design parameters. It allows the engineer to not limit the design to a single deterministic analysis with the most probable parameters (the mode of the distribution), but to evaluate the reliability of the design by considering the dispersion of the design parameters.

Variability is an inherent property of natural materials and rocks or rock masses are no exception to the rule. It arises from the various formation and transformation processes of rock and rock masses, including diagenesis, fractioned crystallisation, alteration through fluid circulation, metamorphism; all of which have local influence on the mechanical parameters of rock and rock masses. Due to this variability, rock mass properties will vary within a rock unit for example along the trace of a tunnel. Thus, a failure mechanism will affect more or less severely various locations along this tunnel.



Here again the PEM approach implemented in *Phase2* is well-suited, and will for example allow the engineer to anticipate what percentage of a tunnel section will be affected by a failure mechanism for a given severity level. It will also allow for an evaluation of the range of severity of a given failure mechanism that should be anticipated and thereby permit the inclusion of flexibility in the design to handle the less probable but potentially more severe situation. Having an estimate of the distribution of the severity of a potential failure mechanism will also permit the optimisation of the support systems and to obtain a better estimation of the cost and thus the economical risk of a project. However, there are some limitations with this approach. Firstly, the severity of the failure mechanism evaluated must be continuous and monotonic: as will be illustrated later in this manual, if abrupt changes in behaviour occur the PEM method can be misleading. Thus, defining domains in which behaviour is similar and the severity of failure is continuous is critical. Finally, the wavelength of the variability must be relatively large compared to the engineering problem evaluated, so that homogeneous domains can be defined. In the case of short wavelength variations, the effect will be different and needs to be treated, not as variability as defined here, but as heterogeneities as discussed in the following.

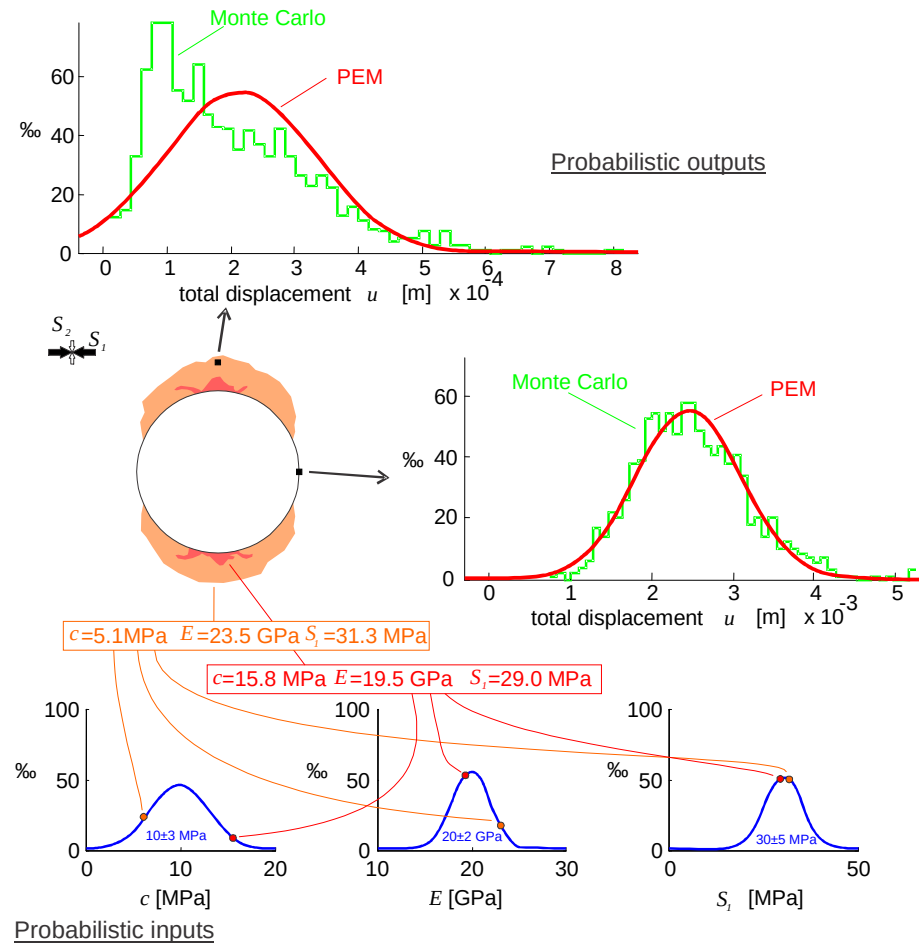
Heterogeneities need to be treated separately as they will not only influence the severity of a failure mechanism, but will also change the behaviour of the rocks or rock masses. For example, increasing modulus heterogeneities in a rock will promote the development of local tensile stress even in an overall compressive field which will affect the failure mode, e.g. change from shear mechanism to tensile dominated mechanism like spalling. The PEM approach implemented in *Phase2* does not simulate the effect of heterogeneities. Heterogeneities must be handled differently, either by the use of a classification system and equivalent homogeneous properties (e.g. defining material properties using GSI) or by explicit modelling of the heterogeneities (e.g. using “Discrete Function” material type).

Limitations

The main limitation of the current implementation of PEM in *Phase2* is that it deals with normal distribution (mean and standard deviation) only. If either input or output random variables differ from a normal distribution they will be approximated by a normal distribution which will lead to inaccuracies.

The user of the PEM implementation must also be aware that even if all the inputs follow normal distributions the outputs may not be normally distributed and the PEM approximation can be inaccurate. This occurs particularly when different behaviours are occurring (e.g. elasticity, plasticity). An example of this situation is presented in the following Figure where a plastic analysis was performed including three random variables all normally distributed: cohesion c , modulus E and maximum far field stress S_I . In parallel to the PEM analyses, a Monte-Carlo analysis was performed to evaluate the probabilistic outputs. The modeled case was a circular opening as displayed in the center part of the Figure with the plastic zone of two realisations (in red and orange) of the Monte-Carlo simulation. The probabilistic outputs presented here are the total displacement at two locations, one in the right wall of the opening and the other some distance in the opening roof. At the location located in the wall, the PEM output is very similar to the distribution obtained with the Monte-Carlo analyses. At this location, the behaviour remains elastic for all the runs (no behaviour change). This is not the case for the point located in the tunnel roof, where mixed behaviour (elastic or plastic) can occur. In this later situation, the distribution obtained from the Monte-Carlo analyses differs significantly from a normal distribution, therefore the PEM output doesn't capture it accurately, though in this case the overall range of the distributions are probably captured, which may not be always the case. However it is noteworthy that if not entirely precise the output of the PEM analyses provides valuable insight in the general shape of the output distribution and this with a computational effort two orders of magnitude smaller than with the Monte-Carlo approach (8 *Phase2* runs for the PEM analyses vs. 1000 *Phase2* runs for the presented Monte-Carlo results).

Clearly, this example illustrates the advantages of the PEM approach but also the caution that must be taken when applying it, particularly when mixed behaviours are occurring. In this latter case the results will probably be inaccurate to some degree.



Recommendations

In light of the material presented above the following recommendations can be made:

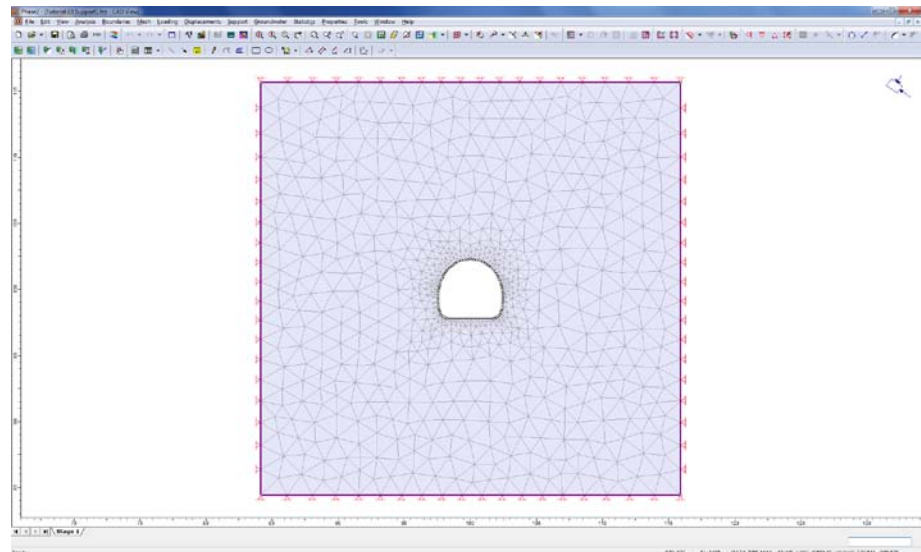
- PEM analyses are applicable for reliability analyses and variability analyses, but don't deal with the effect of heterogeneities.
- A fundamental assumption is that all random variables are normally distributed. When a situation differs from this assumption inaccuracy has to be expected.
- When mixed behaviour occurs (e.g. elasticity/plasticity) one has to expect that the output variables will not be normally distributed and that PEM output will be inaccurate.
- It is recommended to use initially a limited number of random variables and to evaluate their impact on the output

distributions. Keeping the modeling exercise initially simple will help to understand and properly interpret the analyses output.

- It is also recommended to run in addition to a PEM analyses some deterministic extreme cases ($\pm 2\sigma$ or $\pm 3\sigma$) models. Indeed, the evaluation points in a PEM analyses are located at $\pm 1\sigma$ of the mean case and will not capture behaviours that could occur further off the mean.

Model

Start the *Phase2* Model program. For this tutorial, we will start with a model that has already been created. Go to **File** → **Recent Folders** → **Tutorials Folder** and open **Tutorial 03 Support1.fez**. You should see the following model:



This is a model of a tunnel excavated in phyllite with no support. We will assume that there is uncertainty regarding the material properties so we will perform statistical parametric study to observe the possible range of behaviour for this model, using the point estimate method (PEM).

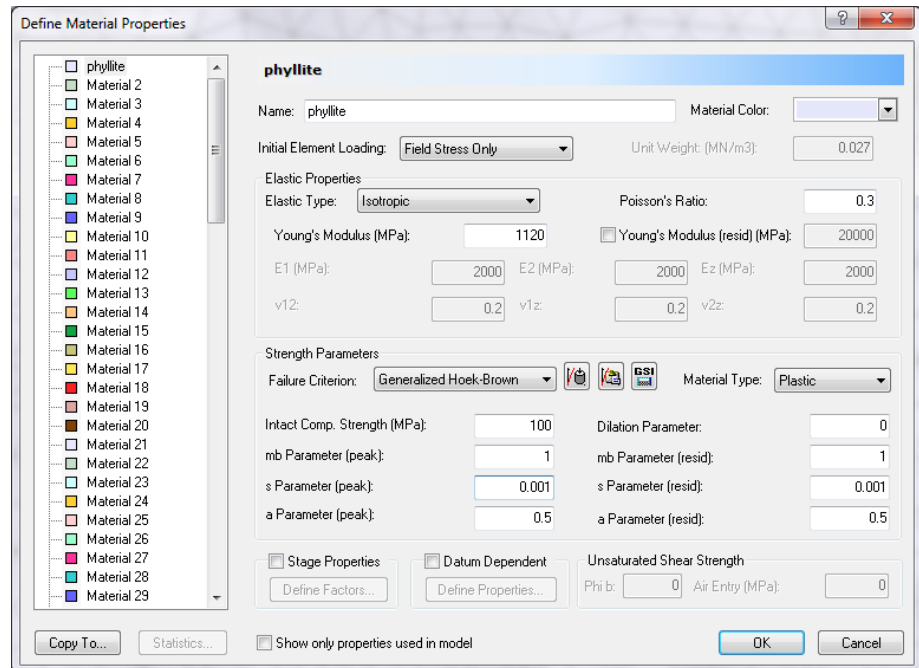
Materials



The material properties have already been assigned for this model. Go to the **Properties** menu and select **Define Materials**. You can see that the phyllite has been assigned a Hoek-Brown failure criterion. Change the failure criterion to Generalized Hoek Brown as shown.

Now we could enter our best guess for the Hoek-Brown parameters and run the analysis. However, the properties of this material are not well known, so we will run a statistical analysis by varying the parameters in a systematic way to see the range of possible behaviours.

This can be done by selecting the **Statistics** button at the bottom of the dialog. You can see that the button is currently grey so we cannot select it. We therefore have to first activate the statistical analysis in *Phase2*.

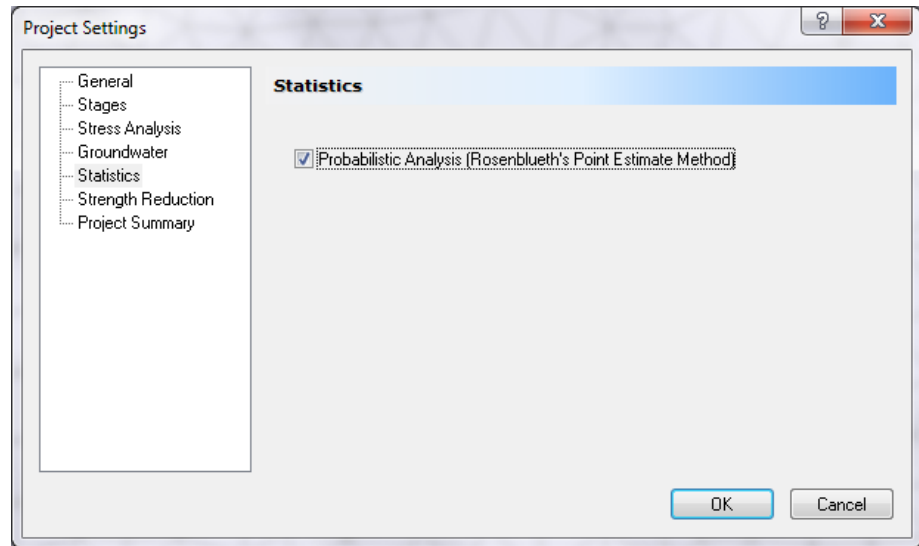


Click OK to close the dialog.

Statistics



To activate the statistical analysis, open the **Project Settings** dialog from the **Analysis** menu and select the **Statistics** page. Select the checkbox for Probabilistic Analysis as shown.



Click OK to close the Project Settings dialog.

We could now go back to the material dialog and set the Generalized Hoek-Brown parameters. Alternatively we can get there through the Statistics menu.

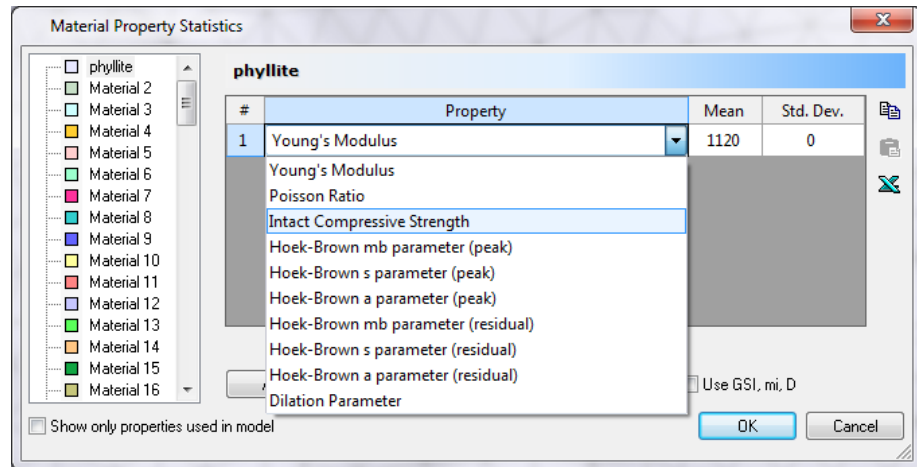
Go to the **Statistics** menu and you will see three options:

- Material Property Statistics
- Joint Property Statistics
- Field Stress Statistics

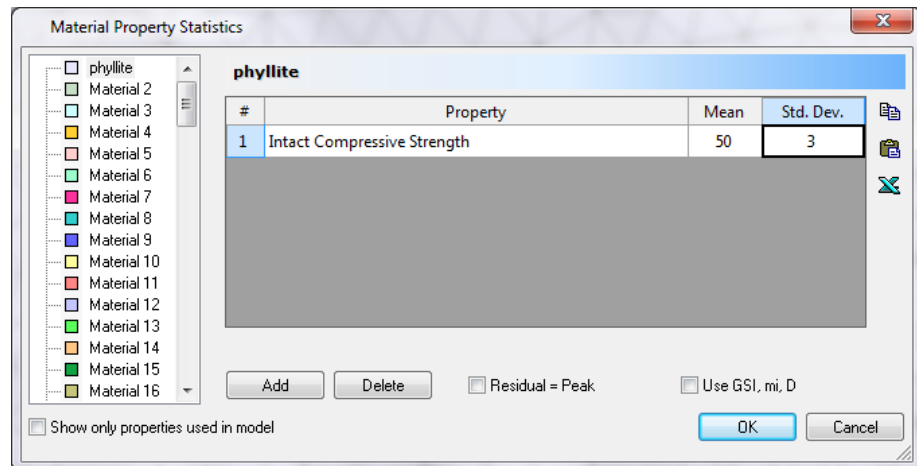
With these options it is possible to perform parametric studies on the input field stress, material properties or joint properties. For this tutorial, we will assume that we know the field stress well, so we will only perform a statistical study on the material properties.

Choose **Material Properties** from the **Statistics** menu. You will now see a dialog which allows you to choose the material property random variables you wish to define for your probabilistic analysis.

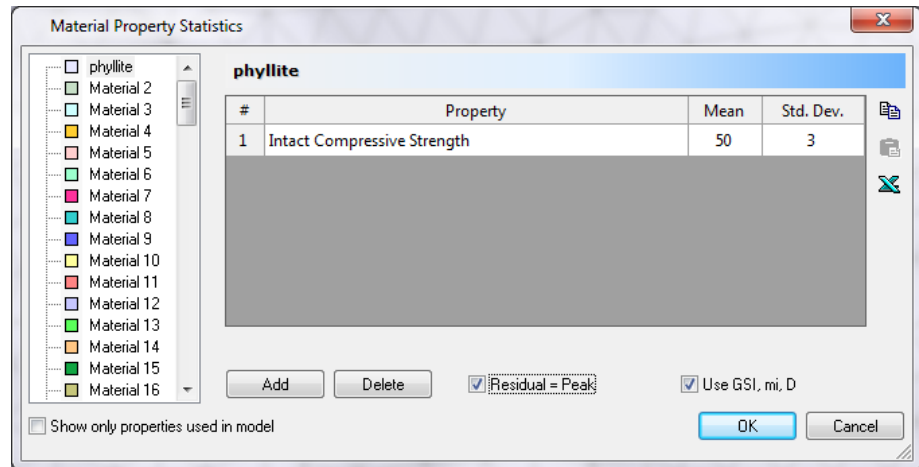
The phyllite material is already selected by default. Click the **Add** button. By default Young's Modulus will be added to the random variable list. You can change this by clicking on the property and selecting the desired parameter from the drop-list of available parameters. For this example, we will choose Intact Compressive Strength as shown.



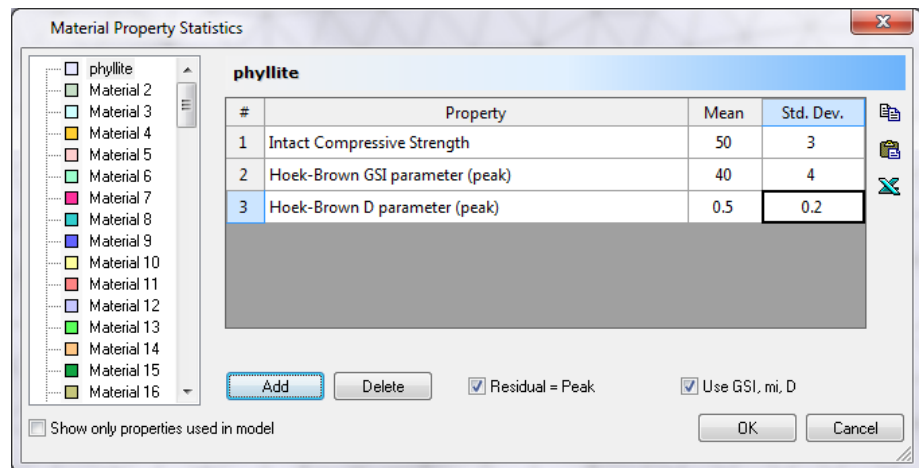
Now we can set the mean (average) and standard deviation for this property. Set the Mean to 50 MPa and the standard deviation to 3 as shown.



Now, let's assume we would rather use GSI, m_i and D to define the Hoek-Brown failure envelope instead of m_b , s and a . Select the **Use GSI, m_i , D** checkbox. We will also assume that the material is perfectly plastic, so we only need to enter the peak values (not residuals). Select the **Residual = Peak** checkbox.



Now let's add two more random variables. Click the Add button twice. For the variable 2, choose Hoek-Brown GSI parameter (peak). Set the mean to 40 and the standard deviation to 4. For variable 3, choose Hoek-Brown D parameter (peak). Set the mean to 0.5 and the standard deviation to 0.2.



Click OK to close the dialog.

NOTE: the number of models that need to be run when testing n random variables is $2^n + 1$. For this example, $n = 3$ therefore $8 + 1 = 9$ separate models will be run. (The 1 additional analysis uses the mean value of each variable and is equivalent to a deterministic *Phase2* analysis). So the user should be aware that testing variations in many parameters simultaneously can generate a very large number of models when using the point estimate method (e.g. 10 random variables would require over 1000 separate analyses).



The model definition is now finished. Save the model using the **Save As** option in the **File** menu. The program actually now saves multiple versions of the file with different material parameters - so it may take a few seconds for the save to complete.

Compute

Run the model using the **Compute** option in the **Analysis** menu.



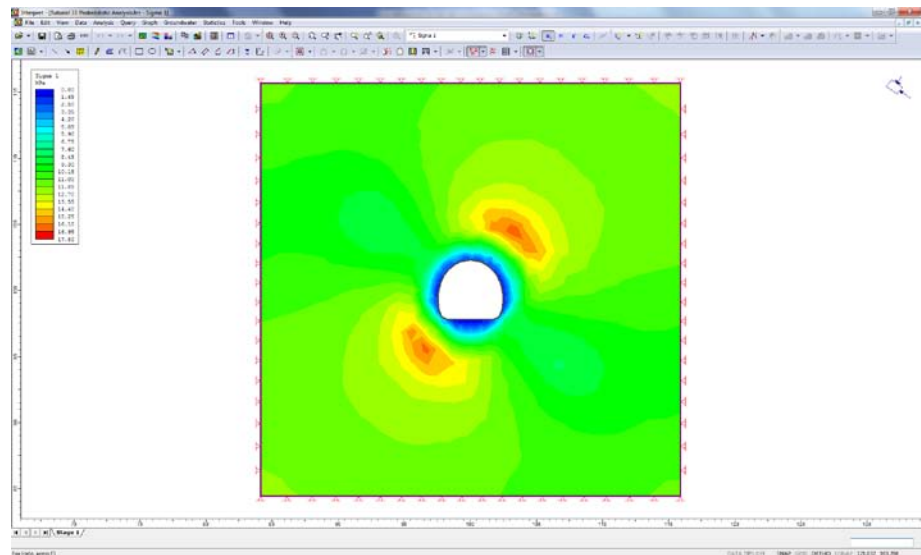
As described above, the statistical analysis is performed by running multiple models with different material properties. If you have a multi-core computer, *Phase2* uses a multi-core parallel processing compute engine by default, so the analysis should not take too long.



Once the model has finished computing select **Interpret** to view the results.

Interpret

The Interpret program starts and reads the results of the probabilistic analysis. You are now looking at the Maximum Compressive Stress (Sigma1).



You can see how there are low stresses around the tunnel where failure has occurred, and high stresses further out.

By default, this plot is showing the model run with the mean value for each parameter. This is called the Base File. Select **Statistics** from the menu and you can choose different plots:

- **Base File Values:** shows the model results when each parameter is set to its mean value.

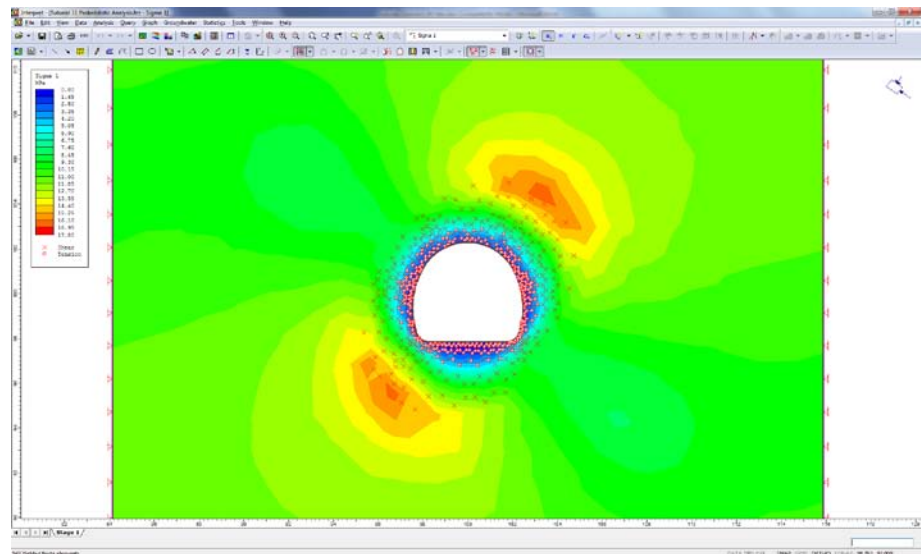
- **Mean Values:** shows the average of all the models
- **Standard Deviation Values:** shows the standard deviation for all the models
- **Coefficient of Variation Values:** shows the ratio of standard deviation to the mean (dimensionless)

Try choosing different options from the **Statistics** menu and observe the range in calculated Sigma1.

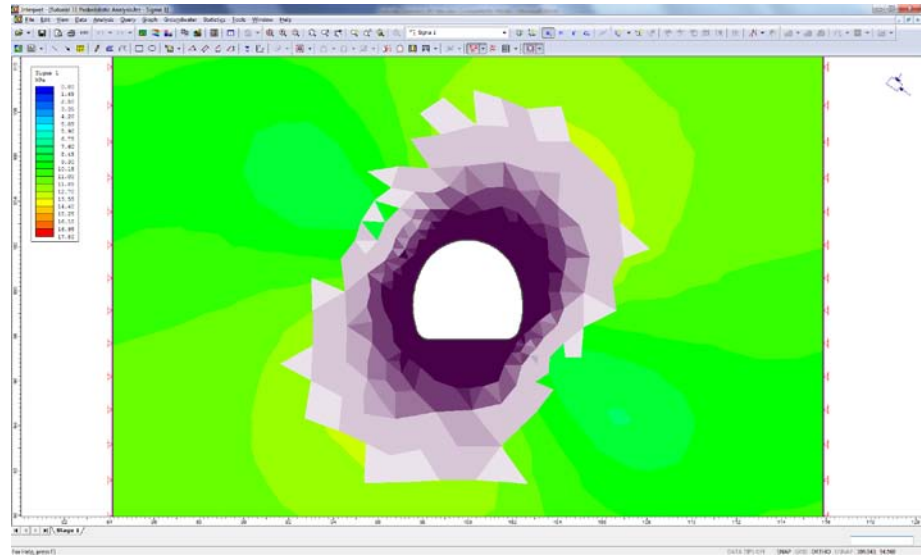
NOTE: the Interpret program calculates the mean and standard deviation 'on the fly' for the different quantities when you plot them. If you have an analysis with many elements and many changing variables, you may notice a significant pause when you change plots. You can solve this by choosing **Compute All Statistics** from the **Statistics** menu. This will calculate everything so that plots will appear instantly.



Go back to showing the base file and show the failed elements using the 'Display Yielded Elements' button. Zoom in on the excavation (using the middle mouse wheel) and the model should look like this:



This shows the failure that occurs in the base file, but perhaps you wish to see the extent of failure as parameters vary. Press the 'Display Yielded Elements' button again to turn off the yielded elements. From the **Statistics** menu, choose **Show Model Yield Zones**. This option will show yielded zones from all component files of the probabilistic analysis. The plot should now look like this:

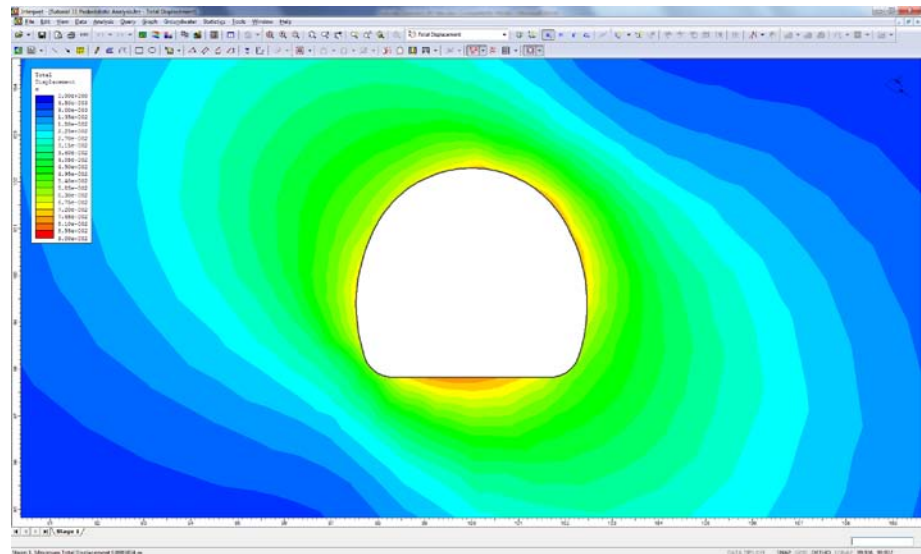


The darkest colour indicates elements that fail in every model. Lighter colours indicate elements that are less likely to fail. This plot is useful when trying to determine the extent of possible failure when designing rock bolt support, for example.

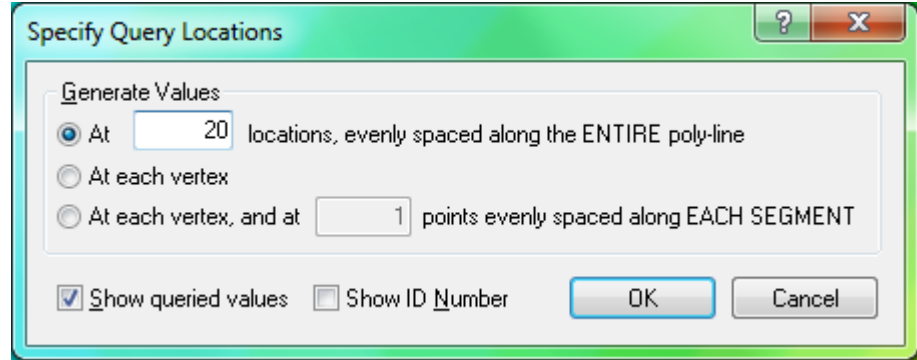
The shading of the yield zones can be customized in the Show Yield Options dialog available in the Statistics menu.

Error plots

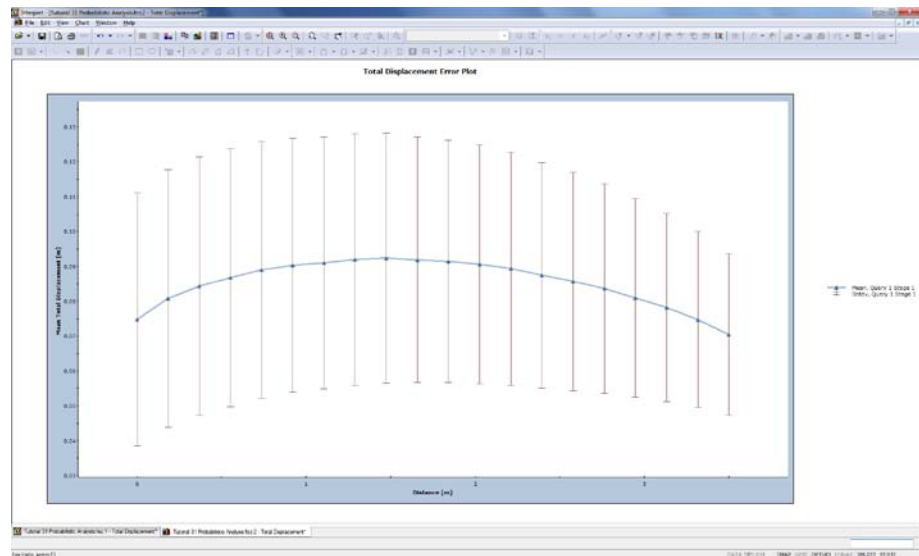
You can also get statistical information for queries. To test this, first turn off the yielded zone display by unselecting **Show Model Yield Zones** from the **Statistics** menu. Now we want to look at the floor heave for this tunnel. Change the plot to show Total Displacement contours.



Go to **Query** → **Add Material Query**. Select a point at the bottom left of the tunnel (the mouse should snap to a point where the bottom flat part starts to curve). Now choose the corresponding point on the bottom right. Hit Enter to finish entering points. You will now see the Query Point dialog. Choose to generate values at 20 locations as shown.



Click OK to close the dialog. You can now generate an error plot for this query. Right click on the query line and select **Error Plot** (or else go to **Statistics** → **Error Plot** and click on the query line). You will now see a plot showing the mean displacement along the bottom of the tunnel with error bars indicating one standard deviation of displacement.



The error plot indicates the range of possible floor displacements that can be expected for the given uncertainty in material properties.

Note: to customize the chart display, right-click on the plot and select **Chart Properties**. Experiment with the various chart properties. You can also zoom and pan the chart display with the mouse wheel.

Close the error plot view.

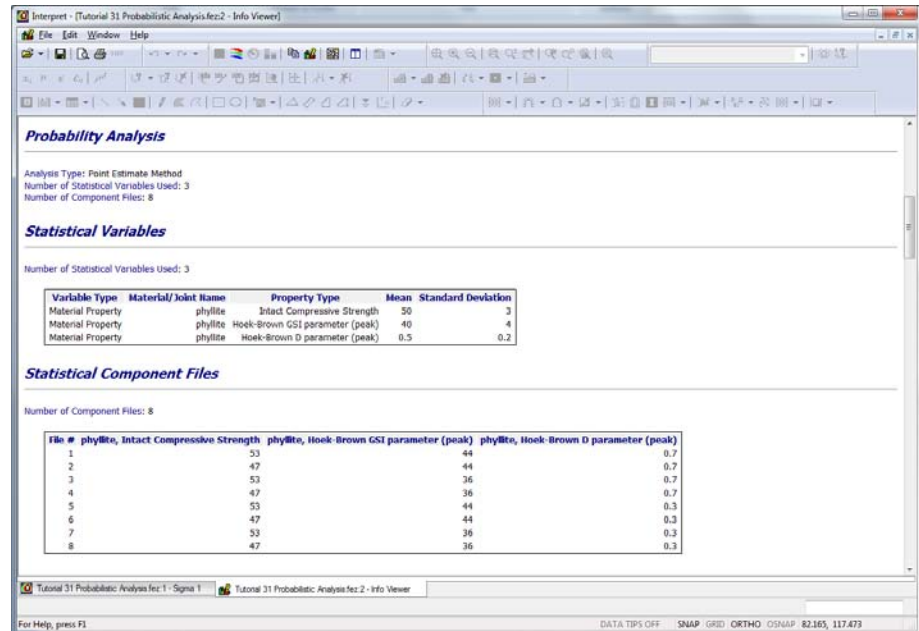
Info Viewer

Finally let's have a look at the Info Viewer.



Select the **Info Viewer** option from the toolbar or the **Analysis** menu.

Scroll down the Info Viewer until you see the probability and statistics sections.



For a probabilistic analysis, the Info Viewer presents a summary of:

- The number of random variables
- The mean and standard deviation of each random variable
- The actual values of the random variables used for each component file of the probabilistic analysis

Since this example uses 3 random variables, there are 8 possible combinations ($2 \times 2 \times 2$) of variables generated by the point estimate method.

This concludes the tutorial. Please exit the *Phase2* Interpret and Model programs.